INFORMATION BRIEF · HEALTH ACCESS

Who lives far from a hospital in Baalbek-Hermel?

Car travel time to the nearest hospital and primary health care centre · Ali Tabikh · October 2026

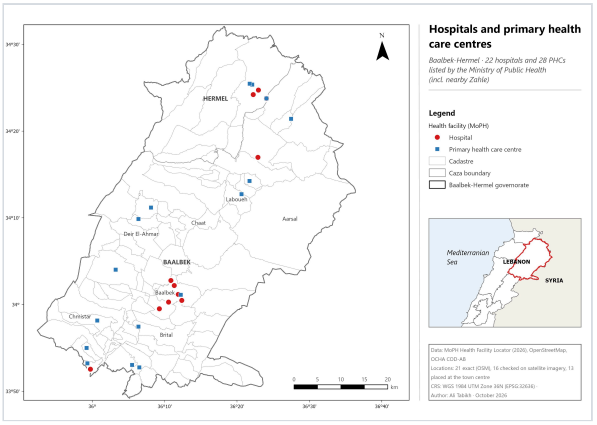
383,000
people in Baalbek-Hermel governorate (WorldPop 2025)

11,100
live more than 30 min by car from a hospital (2.9 %)

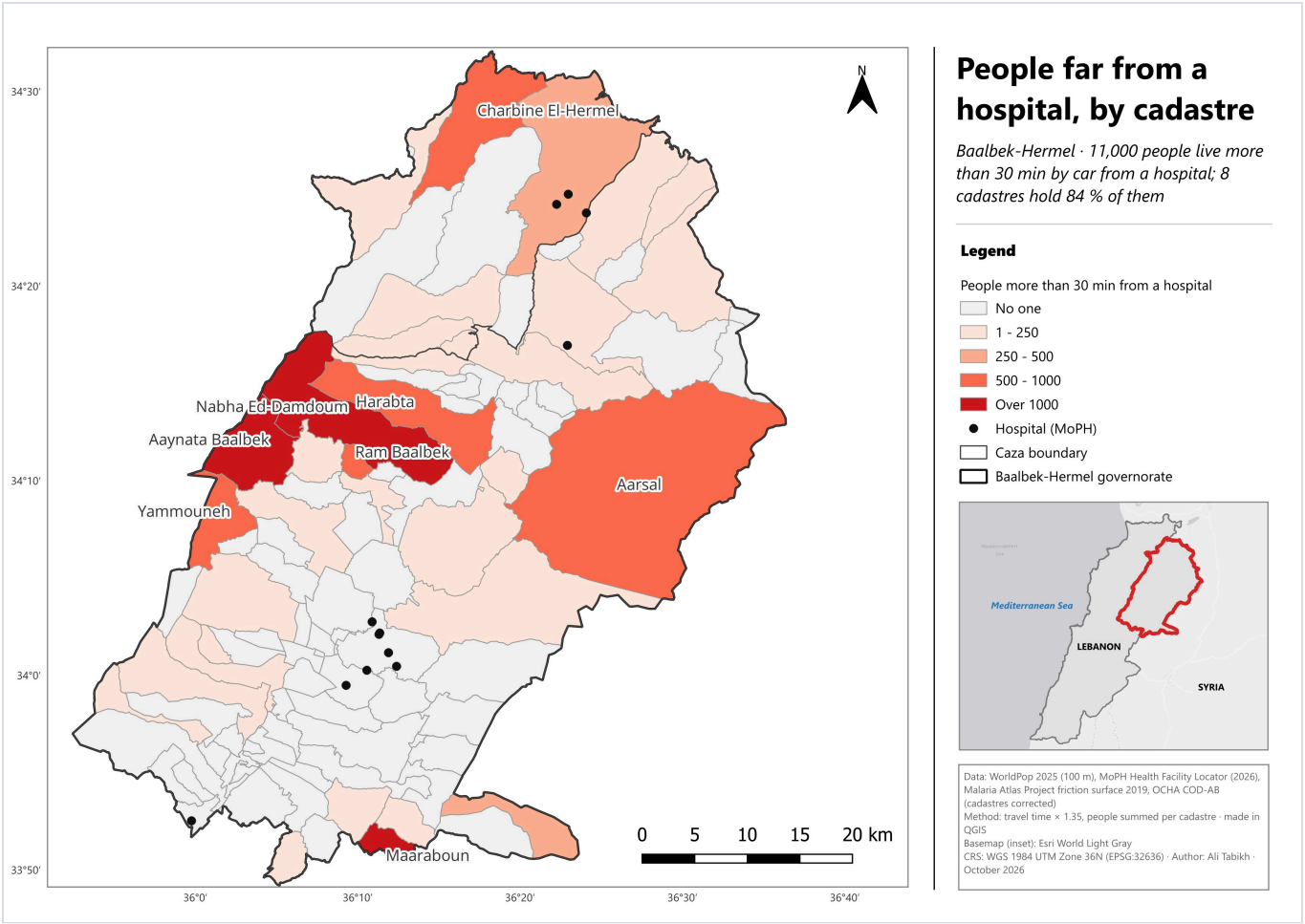
1,900
live more than 30 min from a primary health care centre (0.5 %)

84 %
of the people far from a hospital live in just 8 cadastres

1. **The gap is hospitals, not primary care.** 91 % of people live within 15 min of a primary health care centre (PHC), but only 64 % within 15 min of a hospital.
2. **The people far from a hospital live in two areas:** the western mountain villages (Nabha Ed-Damdoum, Aaynata, Harabta, Ram, Yammouneh) and Maaraboun hold about 7,800 of them; the north-east (Charbine El-Hermel, Aarsal) about 1,500.
3. **In the western villages a PHC is 5–15 min away but a hospital 28–36 min.** Making these PHCs able to stabilise and refer emergencies would reach most of the people at risk.

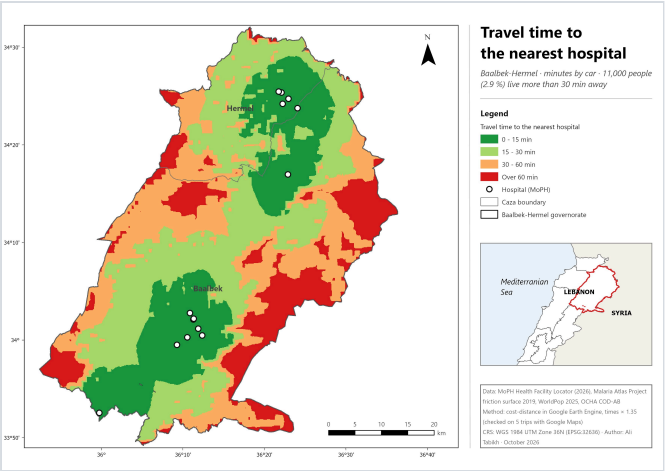


Map 1 · 22 hospitals and 28 PHCs (MoPH), made in ArcGIS Pro.

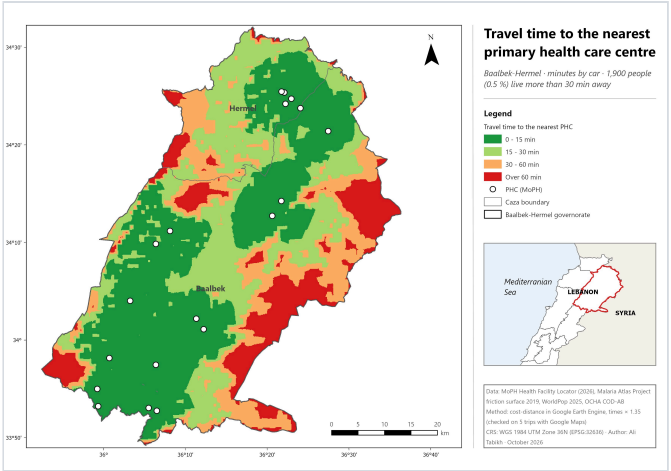


Map 4 · People living more than 30 min by car from a hospital, per cadastre. Made in QGIS.

Findings



Map 2 · Travel time to the nearest hospital (ArcGIS Pro)



Map 3 · Travel time to the nearest PHC: same colours, far more green

Primary care is close for almost everyone. The Ministry of Public Health lists 18 PHCs and 12 hospitals in the two cazas; with the 20 facilities of nearby Zahle, 91 % of people reach a PHC in under 15 min.

Hospitals are grouped in Baalbek and Hermel towns. One person in three (127,000) needs 15–30 min to reach a hospital, and 11,100 need more than 30 min. Baalbek caza holds 9,700 of them, Hermel caza 1,400.

The most isolated place is Tfail (about 300 people, on the Syrian border): about 60 min to a hospital and 50 min to a PHC.

Cadastre	People >30 min	Hospital	PHC
Nabha Ed-Damdoum	2,600	32 min	13 min
Maaraboun	1,700	32 min	21 min
Aaynata Baalbek	1,600	36 min	12 min
Harabta	800	28 min	15 min
Charbine El-Hermel	800	27 min	24 min
Aarsal	700	23 min	14 min
Ram Baalbek	600	32 min	5 min
Yammouneh	500	30 min	14 min

Minutes = average for the people of the cadastre. A cadastre's average can be under 30 min while part of it is further away (Aarsal).

Options for health partners

- **Western mountain villages (about 6,100 people):** equip the nearest PHCs for emergency stabilisation and referral, and agree ambulance cover with the Baalbek hospitals. This is cheaper and faster than a new hospital.
- **Tfail, Charbine El-Hermel and Maaraboun:** here even the nearest PHC is 20 min or more away; a scheduled mobile medical unit fits better than a fixed facility for so few people.
- **Aarsal:** check the real population before planning. WorldPop counts about 25,000 people, likely fewer than live there.

Method

1. **Facilities:** 22 hospitals and 28 PHCs from the MoPH Health Facility Locator (Baalbek, Hermel and Zahle cazas, October 2026), placed from OpenStreetMap (21), satellite imagery (16) or the town centre (13).
2. **Travel time:** Google Earth Engine, motorised friction surface 2019 (Malaria Atlas Project) and cost-distance to the nearest facility. Times × 1.35, the median gap found against Google Maps on 5 real trips.
3. **People:** WorldPop 2025 (100 m) summed per travel band and per cadastre (100 cadastres, OCHA boundaries corrected) in Python; checks with SQL in QGIS; dashboard in Power BI.

Limits

- Travel time is for a car on a 2019 road surface (~1 km); it does not include snow, closed roads or people without a car.
- The × 1.35 factor rests on 5 trips. With raw model times only 1,600 people are over 30 min from a hospital.
- A facility on the list is not a measure of its services, staff or capacity.
- 13 facilities are at the town centre; WorldPop may miss people in informal settlements.

Data: MoPH Health Facility Locator (2026) · OpenStreetMap · Malaria Atlas Project friction surface 2019 (Weiss et al., 2020) · WorldPop 2025 · OCHA COD-AB. Code: GitHub, folder P2_Health_Access.